Crime Records of India

Development Context & Progress Report

Technical Summary of Backend Development Progress

PROJECT OVERVIEW

built using Node.js, Express.js, MySQL, Redis, Docker, and JWT Authentication.

The system supports:

- Dynamic dataset APIs
- Analytics APIs
- Admin governance APIs
- Redis caching layer
- JWT authentication
- Refresh token flow
- Dynamic schema APIs
- Audit logging
- Search and filtering

CURRENT TECH STACK

Backend:

- Node.js
- Express.js

Database:

- MySQL

Caching:

- Redis
- Dockerized Redis container

Authentication:

- JWT Access Tokens
- JWT Refresh Tokens
- Role-based Authentication

Infrastructure:

- Docker Compose
- Redis Container

IMPLEMENTED FEATURES

1. AUTHENTICATION SYSTEM

Completed:

- User Registration
- User Login
- Admin Login
- JWT Access Token generation
- Refresh Token generation
- Profile APIs
- Role-based middleware

Important Notes:

- Admin login uses username
- User login uses email
- Passwords stored using bcrypt hashes

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2. DATASET ENGINE

Completed APIs:

- GET /api/tables
- GET /api/tables/:table/schema
- GET /api/data/:table
- GET /api/data/:table/search
- GET /api/data/:table/filter-options/:column

Features:

- Dynamic table access
- Dynamic schema generation
- Pagination support
- Search functionality
- Filter option generation

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3. ADMIN GOVERNANCE SYSTEM

Completed:

- Update dataset rows
- Delete dataset rows
- Audit logging
- IP tracking
- Change tracking

Database Table:

admin_modification_logs

Tracks:

- Admin username
- Modified table
- Old data
- New data
- Changed columns
- Action type
- IP address
- Timestamp

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4. ANALYTICS SYSTEM

Completed:

- Analytics overview API
- Total users count
- Total admins count
- Total dataset tables
- Total searches
- Total views
- Most viewed dataset
- Recent user logins

API:

GET /api/analytics/overview

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5. REDIS CACHING LAYER

Redis is fully working and integrated.

Cached APIs:

1. GET /api/tables

TTL:

1800 seconds

2. GET /api/tables/:table/schema

TTL:

3600 seconds

3. GET /api/analytics/overview

TTL:

300 seconds

Redis verification was completed successfully.

Example terminal logs:

Fetching tables from MySQL
Serving tables from Redis cache

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6. DOCKER SETUP

Current Decision:
Only Redis runs inside Docker.

Backend:
Runs locally using:
npm run dev

MySQL:
Runs locally

Docker Compose contains only Redis.

Final docker-compose.yml:

version: '3.8'

services:

redis:
image: redis:7

container_name: crime-redis

ports:
- "6379:6379"

volumes:
- redis_data:/data

restart: unless-stopped

volumes:
redis_data:

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7. IMPORTANT FIXES COMPLETED

Fixed Issues:

- Redis connection errors
- responseData undefined errors
- Analytics variable mismatch
- password_hash mismatch issues
- Missing primary key assumptions
- Wrong database import paths
- Port 3000 conflicts
- JWT auth testing issues

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8. DATABASE TABLES

Core Tables:
- users
- admins
- refresh_tokens
- views_log
- search_logs
- admin_modification_logs

Dataset Tables:
- auto_theft
- victims_of_rape
- serious_fraud
- police_housing
- property_stolen_and_recovered
- human_rights_violation_by_police
- trial_of_violent_crimes_by_courts

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9. PROJECT STATUS

Backend Status:

STABLE

Redis Cache:

WORKING

Authentication:

WORKING

Analytics:

WORKING

Governance APIs:

WORKING

Dynamic Dataset APIs:

WORKING

Docker Redis:

WORKING

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10. PENDING FEATURES

Planned Features:
- Rate limiting
- Helmet security
- Input sanitization
- User view history APIs
- Frontend integration
- Deployment configuration
- Swagger/OpenAPI generation
- Production Dockerization

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11. FRONTEND NOTES

Frontend developers should:
- Store JWT access token securely
- Send Authorization header
- Use schema APIs for dynamic rendering
- Use pagination for large datasets
- Handle token expiration properly

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12. IMPORTANT DEVELOPMENT NOTES

Backend start command:

npm run dev

Redis start command:

docker compose up -d

Redis stop command:

docker compose down

Redis container name:

crime-redis

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13. CURRENT PROJECT LEVEL

This project now includes:

- Enterprise backend architecture
- Redis caching
- JWT authentication
- Dynamic APIs
- Audit governance
- Analytics engine
- Docker infrastructure

This is significantly beyond a normal CRUD project.

END OF DEVELOPMENT CONTEXT REPORT